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How generative AI shapes user perceived value and adoption intention in digital museum experiences

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In the context of digital transformation, Generative AI is reshaping cultural heritage dissemination and museum user experiences. This study develops a value-based adoption model to examine how GenAI's adaptability, perceived benefits, and perceived costs influence users' perceived value and adoption intention. Using the British Museum's "The Living Museum" platform, data were collected from 726 Chinese users and analyzed with PLS-SEM. Results show that semantic relevance and contextual adaptability significantly enhance perceived value. Perceived usefulness, enjoyment, novelty, and relative advantage increase perceived value, while complexity and perceived risk reduce it. Service personalization and habit change exerted no significant effects. Perceived value strongly predicts adoption intention, with perceived innovativeness and interactivity moderating this relationship. Multi-group analysis further reveals differences between professional and non-professional users in how novelty and risk affect perceived value. These findings extend value-based adoption theory in digital heritage contexts and provide practical insights for optimizing GenAI-enabled museum services.

In recent years, the rapid advancement of generative artificial intelligence (GenAI) has profoundly shaped the dissemination of cultural heritage. Within museums, GenAI has transcended the limitations of traditional information display, enabling interactive and personalised experiences[1] and accelerating the transition towards the "smart museum" paradigm[2]. For instance, the Metropolitan Museum of Art employs AI-driven dynamic imagery and virtual character interactions to uncover the narratives behind its exhibits (https://bit.ly/4oWEyec, accessed 17 May 2025). The Louvre Abu Dhabi integrates ChatGPT-4 into real-time installations that generate customised allegories (https://bit.ly/4opctMB, accessed 17 May 2025). Similarly, the National Museum of China leverages multilingual AI tour guides to enhance accessibility for non-native speakers (https://themuseumsai.network/toolkit/, accessed 1 June 2025). At the same time, the AI-generated museum tour guide market is expanding rapidly (https://bit.ly/4oqGifI, accessed 8 May 2025). These pioneering practices illustrate the transformative potential of GenAI in advancing cultural heritage dissemination.

Amid these developments, GenAI-driven museum experiences have become a prominent research focus. Existing studies highlight its role in enhancing interactivity[3], improving digital museum comfort[4], enabling artefact detection and personalised services[5], and fostering cultural narrative innovation[6]. Yet its application remains contested. Critics argue that indiscriminate use of AI risks undermining the museum's authority as an educational institution[7]. Model limitations, such as data bias and insufficient cultural understanding, may produce inaccurate, inappropriate, or superficial content[1]. Excessive reliance on AI can also disrupt visitor pacing and reduce direct engagement with exhibits and social interaction[8]. Thus, a binary "good/bad" framework fails to capture the complexity of user evaluations and adoption behaviour. A balanced assessment of both "positive utility" and "potential costs", grounded in technological suitability and user experience, is essential for achieving sustainable integration in cultural heritage contexts.

Although prior studies have addressed these issues, most examine isolated functions or narrow experiential metrics[9]. Research on experiential value tends to emphasise benefits while neglecting the trade-off between effort and reward[10]. User heterogeneity and interactional dynamics within the "value–adoption" relationship are also underestimated[11]. Overall, there remains limited work that systematically evaluates both technological adaptability and perceived user value, as well as the mechanisms shaping user experience and adoption intentions.

To address this gap, the present study develops a comprehensive structural model that integrates technological adaptability (semantic

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relevance (SR) and contextual adaptability (CA)), perceived value (PV) (perceived benefits and perceived costs), behavioural responses (adoption intention), and key moderating factors (perceived innovativeness and interactivity). Drawing on user data from the British Museum's The Living Museum platform, and employing factor analysis and structural equation modelling in SPSS 27 and SmartPLS 4, this study investigates four core questions:

RQ1: How does the technological adaptability of GenAI influence users' PV of museum services?

RQ2: How do users' perceptions of benefits and costs shape overall value assessments?

RQ3: Does PV drive adoption intent, and are these effects moderated by perceived innovativeness and interactivity?

RQ4: Do adoption mechanisms differ significantly across user groups?

This study proposes a GenAI user analysis framework tailored to cultural heritage contexts. The framework identifies the critical factors and underlying logic that shape interactive museum experiences and adoption behaviors. It also provides a theoretical foundation and practical guidance for optimising AI-driven cultural services and design strategies.

Methods

Adaptability of GenAI in museums

GenAI is commonly defined as a technology that utilizes deep learning models to generate human-like content—such as text or images—in response to complex and varied prompts, including natural language, instructions, or questions[12]. As digital technologies become increasingly integrated into cultural institutions, GenAI is widely recognized as a transformative tool capable of enhancing operational efficiency, visitor engagement, and the dissemination of cultural knowledge within museum settings. This recognition stems from its dual capacity for content generation and intelligent interaction[13]. According to French and Villaespesa, current applications of AI in museums can be broadly categorized into three core domains[1]. First, in the realm of collection management and cultural heritage digitization, museums are leveraging computer vision and image recognition technologies to streamline the processing of collection data. These tools enable the automated extraction of object attributes and support the visualization of cross-dimensional metadata[14]. Second, AI facilitates behavioral analytics by aggregating and interpreting visitor data across pre-visit, on-site, and post-visit stages. Through techniques, such as emotion recognition and behavior analysis, museums can assess visitor satisfaction and engagement with specific exhibitions[15], while also identifying high-frequency participants and at-risk user segments. Third, GenAI plays a pivotal role in advancing voice interaction systems. Thanks to advancements in natural language processing, GenAI-powered guides now support real-time speech recognition and semantic interpretation, enabling multilingual, dynamic conversations. Visitors can navigate exhibitions using voice commands, customizing their tour paths at a self-directed pace[16].

However, it is essential to acknowledge that GenAI operates as a large-scale modeling system—one designed, trained, and validated by human developers. As such, its internal mechanisms inevitably reflect embedded human biases. GenAI should not be regarded as a 'one size fits all' solution[17]. Without appropriate guidelines and expert oversight, the uncritical deployment of GenAI may compromise the editorial integrity and epistemic authority of museums. This can inflate visitor expectations and distort perceptions of the depth and reliability of AI-generated content[1], ultimately undermining the quality of cultural knowledge dissemination. Moreover, the ethical risks associated with GenAI—such as bias propagation, limited model robustness, and potentially harmful or misleading outputs—have become a growing focus of academic scrutiny[18]. To ensure responsible development and implementation, GenAI must be evaluated across multiple dimensions, including its technical underpinnings, institutional governance, and broader social acceptability.

From a semantic perspective, Moyano et al.[19] investigated the potential of large language models (LLMs) to autonomously identify, map, and align equivalent attributes across heterogeneous semantic ontologies and controlled vocabularies. Their research addresses a key challenge in cultural heritage informatics: how to effectively align existing metadata structures with general-purpose knowledge frameworks to enable cross-domain semantic interoperability. Achieving such interoperability would significantly improve the consistency and scalability of GenAI in constructing multi-domain cultural knowledge. It would also provide the theoretical and technical foundation for integrating AI-generated content with domain-specific cultural semantics in a structured manner. López-Martínez et al.[20] proposed a low-maintenance, gamified intelligent guide system for museums that applies semantic web technologies and linked data in innovative ways. The system uses knowledge graphs to semantically model exhibit attributes and employs rule-based engines to automatically generate interactive question-and-answer tasks. This approach enhances the depth of interaction between visitors and exhibits, promoting more immersive and meaningful museum experiences. In parallel, GenAI's capacity to perceive and respond dynamically to visitor contexts has emerged as a key research focus. Nawara et al. [21] proposed a 'loosely coupled' model that synchronizes LLM-based chatbots with users' real-time browsing paths. This system invokes external LLMs at the page level to generate highly relevant questions and personalized content recommendations based on the user's immediate context. Empirical results indicate that this adaptive mechanism enhances visitors' cognitive engagement and increases their intention to explore exhibitions more thoroughly. However, the model's effectiveness is constrained by its reliance on static knowledge bases, limiting its flexibility in addressing diverse user needs. To overcome these limitations, Varitimiadis et al.[22] introduced a distributed semantic reasoning framework designed for multi-agent dialogue systems. Built upon knowledge graph infrastructure and context-aware computing, this framework enables AI systems to dynamically adapt to users' cognitive states and interaction histories. It supports real-time optimization of dialogue strategies and content delivery, thereby enhancing user interactivity. This approach allows GenAI to facilitate responsive, context-sensitive knowledge dissemination and personalized recommendations in museum environments.

Building upon these theoretical and technological foundations (see Table 1), the present study investigates the adaptive performance of GenAI within museum contexts. It focuses specifically on how the dimensions of semantic relevance (SR) and contextual adaptability (CA) influence users' PV and their intention to adopt GenAI-based services.

Perceived Value theory

The concept of 'value' has long served as a cornerstone in economic theory, with its origins grounded in classical frameworks, such as exchange theory, utility theory, and the labor theory of value. Over time, this foundational concept has been enriched by insights from psychology and social psychology, which have contributed to a more multidimensional understanding of value in the context of human behavior[23]. Within the field of marketing, perceived value (PV) has emerged as a critical construct for explaining consumer decision-making. It applies across a wide range of contexts, including physical products, intangible services, and hybrid experiential offerings[24]. Zeithaml[25] defined PV as the consumer's overall evaluation of the utility of a product or service, based on a subjective comparison of 'what is received' versus 'what is given'. This definition closely aligns with the dual concepts of 'perceived benefits' and 'perceived sacrifices' proposed by Dodds et al.[26], as well as the notions of 'desired attributes' and 'attributes of sacrifice' articulated by Woodruff and Gardial[27]. Collectively, these frameworks emphasize the complex, integrative nature of PV, which results from multiple interacting evaluative dimensions rather than a simple, linear trade-off.

Initial research on PV largely focused on two primary dimensions—perceived quality and monetary price[28]. However, as consumer behavior scholarship advanced, a broader understanding of value emerged. Scholars now acknowledge that PV extends beyond functional utility[23], encompassing additional non-functional dimensions, such as social, emotional, epistemic, and conditional value[29]. These factors shape the holistic consumer experience and influence behavioral intentions. In line with this expanded perspective, Kim et al.[23] introduced the Value-Based Adoption

Table 1 | Research on the Adaptability and Usage Pathways of GenAI in Museums

Author	Research subject	Main technology	Research contribution
Casillo et al.[98]	Cultural Heritage in Archaeological Park	Ontology-Driven Chatbots, QA Systems	Ontology-Based Personalized Dialogue for Cultural Tourism Chatbots
Réby et al.[99]	Italian Cultural Heritage	Semantic Web, Linked Data, Knowledge Graph	Italian Cultural Heritage Knowledge Graph for Semantic Linking and Interoperability
Varitimiadis et al.[22]	Museum Chatbots	Graph-Based MAS, Knowledge Graphs, Distributed Collaborative AI	GPT-4 and Knowledge Graph Based Multi-Agent Dialogue Framework
Song et al.[100]	Informal Learning Museum Visitors	AI-Based Recommendation Systems	Impact of Recommender Systems on Heritage Aesthetic Experience
Li et al.[13]	Museum Exhibitions	AI, Immersive Tech, IoT, Cloud Computing	Review of DTT Applications in Museums with AI/VR/AR/IoT Integration
Izadi and Forouzanfar[101]	Intelligent Chatbots	Error Correction, RL/Supervised/Unsupervised Learning, GANs, XAI	Chatbot Error Types, Correction Mechanisms, and Ethical Challenges
Trichopoulos et al.[102]	Museum Visitors	ChatGPT-4 LLM	GPT-4 in Museum Tours with Human-AI Verification Mechanism
Trichopoulos et al.[103]	Museum Visitors	GPT-4 LLM with Context-Aware Fine-Tuning	LLM-Based Personalized Recommendation in Museums
Tsepapadakis et al.[104]	Heritage Site Visitors	Audio AR, Voice Dialogue Systems, IoT	AR, 3D Audio, and Conversational AI for Immersive Audio Tours

Model (VAM), which explains users' adoption of emerging technologies through the lens of value maximization. VAM conceptualizes PV as comprising four key components: Usefulness (extrinsic and cognitive benefits), Enjoyment (intrinsic and affective rewards), Technicality (non-monetary sacrifices), and Perceived Fee (monetary sacrifices). The model posits that users evaluate these dimensions collectively, forming an overall perception of value that informs their decisions to adopt.

As technological innovation accelerates, the modes through which new products and services are delivered have become increasingly diverse. However, user adoption patterns remain inconsistent. To address this discrepancy, Souza et al.[30] sought to investigate the mechanisms underlying the formation of PV and its influence on adoption intentions. Drawing from several prominent theoretical models—including VAM, the Technology Acceptance Model (TAM), Innovation Diffusion Theory (IDT), the Unified Theory of Acceptance and Use of Technology (UTAUT), and the Consumer Acceptance of Technology model (CAT)—they constructed a comprehensive framework incorporating variables, such as perceived usefulness, perceived ease of use, relative advantage, compatibility, complexity, facilitating conditions, pleasure, perceived security, and perceived cost. Their findings confirmed the significant role these variables play in shaping PV and further demonstrated that PV acts as a mediator in determining users' behavioral intention to adopt new technologies.

In summary, as emerging technologies continue to permeate fields, such as culture, commerce, and education, scholars have increasingly focused on identifying the antecedent factors that shape PV and the mechanisms by which this value influences adoption behavior. This line of inquiry has evolved into a multidimensional research paradigm (see Table 2). The present study extends this line of research by examining the use of GenAI in museum contexts. A theoretical model is proposed that incorporates both benefit-related and cost-related factors. The benefit dimension includes perceived usefulness, perceived enjoyment, perceived novelty, relative advantage, and service personalisation. The cost dimension captures perceived risk, complexity, and habit change—key inhibitors to user acceptance. Furthermore, perceived innovativeness and interactivity are introduced as moderating variables. The model is designed to explore how these variables jointly influence users' overall value assessments and their intention to adopt GenAI technologies in cultural heritage settings.

Research model and hypothesis development

This study examines the adaptability of GenAI, with a specific focus on its capacity to interpret and respond to diverse semantic inputs and varying contextual conditions. Semantic relevance (SR) is defined as the degree to which generated information or recommendations closely align with a user's query or underlying needs. It extends beyond surface-level keyword matching to emphasize deeper semantic consistency across information units[31]. Contextual adaptability (CA), by contrast, refers to an AI system's capacity to dynamically adjust the presentation of content and behavioral responses based on specific usage scenarios, user preferences, and environmental variables[32].

A growing body of research confirms that user experience quality significantly improves when AI-generated content is closely aligned with users' expectations and contextual needs. Conversely, when output content is misaligned with the user's context, it is often perceived as intrusive or irrelevant, thereby reducing its PV[10]. For example, in cultural product design, AI-generated outputs that diverge from the intended theme can diminish aesthetic appeal and hinder user acceptance[33]. Similarly, CA has been shown to significantly enhance PV. Bai and Yang[34] found that tailoring GenAI-generated content to suit different disciplinary and educational contexts markedly improved both its instrumental utility and PV. In the domain of museum applications, Ivanov[35] demonstrated that AI guide systems capable of real-time adaptation—based on visitor location and interaction history—enhanced informational relevance and the efficiency of knowledge transmission. Informed by this theoretical and empirical evidence, the following hypotheses are proposed:

H1. SR of GenAI positively influences users' PV.

H2. CA of GenAI positively influences users' PV.

Perceived Usefulness (PU) is defined as the degree to which users believe that employing a particular system enhances their task performance[36]. Within the motivational framework of the TAM, PU represents users' outcome expectations and extrinsic motivation, serving as a critical determinant of technology adoption behavior[37]. Extensive empirical research has consistently confirmed the positive association between PU and users' attitudes toward, and willingness to engage with, new technologies in the domain of cultural heritage[38,39]. In the specific context of GenAI applications in museums, PU refers to users' evaluation of the system's capacity to deliver efficient, accurate, and trustworthy information related to cultural heritage content. Given that GenAI functions as an interactive service, users often assess its utility based on real-world usage and experience[40]. Accordingly, PU is regarded as a key component in the formation of users' PV. Based on the above analysis, the following hypothesis is proposed:

H3. PU positively influences users' PV.

Perceived Enjoyment (PE) refers to the intrinsic pleasure users derive from engaging with a system or platform, independent of any instrumental

Table 2 | Main factors influencing PV and behavioural intention

Author	Context of the study	Theoretical framework	Key factors
Garrouch and Mzoughi[105]	Mobile Shopping Behavior by User Type	SOR	Relative advantage, Complexity, Perceived security, Uncertainty avoidance, Collectivism
Hameed et al.[106]	Mobile Payment System Adoption	VAM	Perceived controllability, Susceptibility to threat, Severity of Threat, Hedonic value, Utilitarian value, Social Influence, Intentions to use, Word-of-mouth
Fatima et al.[107]	Factors Influencing M-Payment Adoption	UTAUT2	Performance expectancy, Effort expectancy, Social Influence, Facilitating conditions, Hedonic Motivation, Price Value, Habit
Jiang et al.[108]	Mobile Digital Music Choice Factors	VAM	Perceived price, Perceived quality, Social value, Emotional value, Conditional value, Epistemic value
Jan et al.[109]	(De)Motivators for Using AI Shopping Agents	BRT	Motivators of technology readines, Inhibitors of technology readiness, Reasons for usage, Reasons against usage, Attitude towards usage
Skandali et al.[110]	AI Service Adoption in Luxury Resorts	VAM	Happiness, Perceived immersion, Trust, Changes in habits
Seon et al.[111]	Factors Affecting Continued Use of GenAI	VAM	Perceived usefulness, Interactivity, Perceived cost, Privacy security risks, Information provision, Service personalization
Xu et al.[112]	Continuance Intentions for AI Painting Tools	ECM, TAM	Perceived Intelligence, Perceived novelty, Perceived trust, Confirmation of Expectations, Satisfaction, Perceived Usefulness, Attitude, Emotion Experience Interactive Experience
Xu et al.[93]	Gen MZ Response to AI Brand Endorsers	CAB	Attractiveness, Anthropomorphism, Interactivity, Authenticity, Hedonic motivation, Social presence, Trust in AI Brand Endorsers, Technology readiness (Optimism, Innovativeness)
Nguyen[113]	Generative AI Use in Academic Research	IDT	Relative advantage, Compatibility, Complexity, Trialability, Observability, Perceived risks, Perceived price value, Characteristics of Communication channels, AI adoption

or performance-related outcomes it may offer[23]. Davis et al.[41] demonstrated that users are more likely to adopt a technology and use it more frequently when they experience immediate enjoyment or find the usage process inherently pleasurable, rather than merely functional. For example, Wan and Jiang[42] found that emotional bonding and prosocial interactions between AI anchors and users during two-way communication significantly enhance hedonic experiences and strengthen users' positive behavioral intentions. Similarly, Shao[43] showed that the hedonic value generated by AI-powered virtual anchors exerts a substantial positive effect on customer engagement and brand perception. Based on these findings, the following hypothesis is proposed:

H4. PE positively influences users' PV.

Perceived Novelty (PN) refers to users' subjective perception of the innovativeness and uniqueness of a given technology or service. Users are more likely to experience a sense of excitement and curiosity when they perceive a technology to exhibit distinctive functional features or novel experiential elements. Compared with familiar technologies, novel systems tend to elicit stronger perceptual stimuli, which can, in turn, stimulate exploratory motivation and emotional engagement[44]. Andreassen and Streukens[45] argue that the pursuit of novelty reflects users' openness to technological innovation and their willingness to experiment with unfamiliar solutions. As such, PN serves as a critical driver of new technology and service adoption. Moreover, it has been identified as an important dimension contributing to users' PV[46]. Existing literature suggests that users' PV tends to increase when a technology or service is seen as novel, useful, or capable of meeting emerging needs[47]. Within the context of this study, we posit that when users experience a heightened sense of novelty while interacting with GenAI technologies in museum settings, their PV of the system is likely to increase. Accordingly, the following hypothesis is proposed:

H5. PN positively influences users' PV.

Relative Advantage (RA) refers to the extent to which individuals perceive a new technology as superior to existing alternatives in terms of functionality, efficiency, or overall performance[48]. Extensive empirical research has identified RA as a critical determinant of users' willingness to adopt new technologies. Lin et al.[49] further emphasized that RA significantly influences technology adoption behavior and serves as a key positive driver of PV. Nonetheless, the perceived benefits of a technology are often weighed against its associated costs. For instance, Ayanwale and Ndlovu[50] found that while university students generally acknowledge the practicality of AI tools —such as chatbots—for educational purposes, their adoption intentions may be moderated by perceived effort or risk. In this context, RA captures users' belief that a given technology or service offers superior task performance compared to traditional systems. This task-oriented evaluation directly responds to users' functional needs[51]. Applying this concept to the museum setting, the current study posits that if visitors perceive GenAI technologies as offering greater relative advantage over conventional exhibition systems, they are more likely to recognize their value and adopt them. Based on this rationale, the following hypothesis is proposed:

H6. RA positively influences users' PV.

Service personalisation (SP) refers to the delivery of tailored services based on users' behavioral characteristics, interests, preferences, and historical interaction data[52]. Among the various factors influencing PV, SP has been identified as one of the most influential. Chan and Lee[53] found that members of Generation Z express a clear preference for personalized technologies and show positive attitudes toward the potential of GenAI to improve productivity, efficiency, and individualised experiences. Empirical studies further support this perspective. Wang et al.[54] demonstrated that personalization in smartphone applications has a significant impact on PV. Similarly, Akdim and Casaló[10], in their investigation of voice assistants, reported that personalised recommendations considerably enhance users' perception of service value. In the context of GenAI applications in museums, the implementation of personalised features, such as custom-guided tours, preference-based navigation, and adaptive responses to user behavior, can create more immersive experiences and elevate users' overall value perception. Based on this reasoning, the following hypothesis is proposed:

H7. SP positively influences users' PV.

Habit Change (HC) refers to a substantial deviation from individuals' established behavioral routines or technological usage patterns[55]. In the context of information technology use, HC is often manifested when users perform technological tasks automatically based on prior experience, or when they develop behavioral path dependencies that operate without deliberate cognitive engagement[56]. Empirical findings in information

systems research suggest that habitual behaviors exert a significant influence on users' emotional attitudes and strongly shape their actual usage patterns. For example, in mobile device contexts, habits have been identified as a major predictor of continued use[57]. Moreover, Wathieu[58] argues that repetitive behavioral engagement can reinforce the PV of a service. Within this framework, the present study conceptualizes habit change as a form of cognitive and behavioral cost that users must incur when adopting GenAI technologies. Based on this interpretation, the following hypothesis is proposed:

H8. HC negatively influences users' PV.

Complexity (CO) is defined as the extent to which users perceive a new technology as difficult to understand or operate[59]. In the field of technology adoption research, complexity has been consistently identified as a key factor influencing user behavior[60]. In contrast, technologies that are easy to learn and use, requiring minimal cognitive or behavioral effort, tend to experience more rapid diffusion and broader acceptance[48]. Moreover, CO constitutes a central dimension of users' perceived effort and is closely associated with their evaluation of PV[61]. When a system is perceived as overly complex, users are more likely to view it as burdensome or inefficient, which can diminish the value they attribute to the technology. Accordingly, this study hypothesizes that the perceived complexity of GenAI applications in museum settings negatively affects users' value judgments. The following hypothesis is proposed:

H9. CO negatively influences users' PV.

Perceived Risk (PR) refers to an individual's subjective belief that the use of a given technology may result in negative consequences. This perception typically encompasses both uncertainty and the potential for loss[62]. In the context of GenAI applications within museums, PR may arise from concerns related to data privacy, information security, and the potential erosion of users' critical thinking and analytical skills[63]. However, research findings on the behavioral consequences of PR remain mixed. For instance, Molinillo et al.[64] in their study on AI services, observed that while privacy-related risks significantly affect users' PV, they do not always exert a statistically significant influence on usage intention. Drawing on this literature, the current study posits the following hypothesis:

H10. PR negatively influences users' PV.

From the user's perspective, value acquisition is a central objective of a successful consumption experience[65]. Schechter et al.[66] contend that PV represents a comprehensive evaluation that integrates both quantitative and qualitative, as well as objective and subjective, considerations. It reflects the user's trade-off between the benefits received and the costs incurred during product or service use. Users' perception of value can be enhanced either by increasing perceived benefits or by reducing perceived costs[67]. A substantial body of literature confirms that PV significantly influences consumer attitudes and behavioral intentions[68]. Empirical studies in digital environments further demonstrate that PV positively predicts users' intentions to engage with specific technologies[69]. Based on these findings, the following hypothesis is proposed:

H11. Users' PV positively influences their intention to adopt GenAI technology.

In the field of marketing, perceived innovativeness (PI) refers to an individual's propensity to seek out and adopt new products or experiences, rather than relying on established routines[70]. To maximize PV, it is essential to satisfy consumers' preference for novelty by offering innovations that deliver enhanced utility[71]. Prior research has shown that in AI-enabled virtual tourism environments, traits, such as optimism and innovativeness significantly and positively moderate the relationship between consumer well-being and behavioral intention[72]. Conversely, Shahbaz et al[73] found that innovation resistance negatively moderates the link between behavioral intention and actual behavior in the context of green supply chain management and big data analytics. Extending these insights to the application of GenAI in museums, this study proposes the following hypothesis:

H12. PI positively moderates the relationship between PV and AI.

Interactivity (IN) refers to the exchange of information and dynamic feedback that occurs between individuals, groups, or systems during the diffusion of new technologies within a society or organization[74]. According to the 'Computers Are Social Actors' (CASA) paradigm, interactions between humans and technological systems can be interpreted as instances of human-like behavior. Within this framework, technologies emulate social norms to respond to users in a context-aware manner, thereby enhancing perceptions of social presence and interactive realism[75]. In GenAI-related research, Faust et al.[76] emphasized the pivotal role of interactivity in both system implementation and user evaluation, arguing that higher levels of interactivity significantly strengthen users' intention to adopt the technology. Similarly, De Angeli et al.[77] found that highly interactive interfaces have a positive impact on user experience and overall preferences. Based on this evidence, the present study proposes the following hypothesis:

H13. IN positively moderates the relationship between PV and AI.

Proposed research model

This study extends the Value-Based Adoption Model (VAM) by integrating a set of research hypotheses into a unified theoretical framework (see Fig. 1). The model aims to investigate the causal relationship between museum visitors' overall PV of GenAI and their intention to adopt the technology. In this framework, GenAI adaptability refers to the system's capacity to interpret and respond to diverse semantic content and contextual environments. The constructs of perceived benefits and perceived costs encompass, respectively, the positive outcomes associated with GenAI, such as usefulness, enjoyment, novelty, relative advantage, and service personalisation, and the potential drawbacks, including habit Change, system complexity, and perceived risks. PV is defined as the user's holistic evaluation of GenAI's utility, while adoption intention reflects the individual's subjective willingness to use GenAI services within the museum setting. Furthermore, the model introduces perceived innovativeness and interactivity as moderating variables. These factors represent users' sensitivity to the novelty of technological features and their perception of dynamic human–technology interaction. The model explores how these moderators amplify the relationship between PV and the intention to adopt GenAI.

Results

Questionnaire Development

This study adopts a quantitative survey methodology to collect empirical data, with the questionnaire comprising three main sections. The first section aims to familiarize respondents with the GenAI application scenario addressed in the study, offering essential background on the technological context prior to engaging with the survey items. For this purpose, The Living Museum, an experimental digital platform developed by independent AI engineer Jonathan Talmi and hosted by the British Museum, was selected as the reference case (https://livingmuseum.app, accessed 1 June 2025). This platform integrates GenAI with cultural heritage databases to create an interactive online experience. Users can engage with the system through natural language input, allowing them to filter and explore museum objects based on attributes, such as curatorial department, historical or cultural period, object type, and place of origin. By simulating a dialogue with collection items, the platform facilitates deeper exploration of their historical and cultural significance. Moreover, users can curate personalized thematic collections by freely combining artifacts according to their interests, thereby experiencing a form of individualized cultural curation. The GenAI application scenario represented by The Living Museum is employed in this study to simulate user interaction with GenAI in a digital museum environment. Figure 2, Fig. 3 illustrates the platform's user interface and core functionalities, providing a standardized reference framework for the subsequent measurement constructs in the survey.

The second section collects respondents' demographic information, including gender, age, education level, occupation, and reason for visiting. These data facilitate analysis of potential differences in technology adoption intentions across diverse user groups. The third section comprises 14 latent variables, each measured through a total of 50 items using a 5-point Likert scale (1 = 'Strongly Disagree'; 5 = 'Strongly Agree'). To ensure the scientific validity of the instrument, all measurement items were adapted from

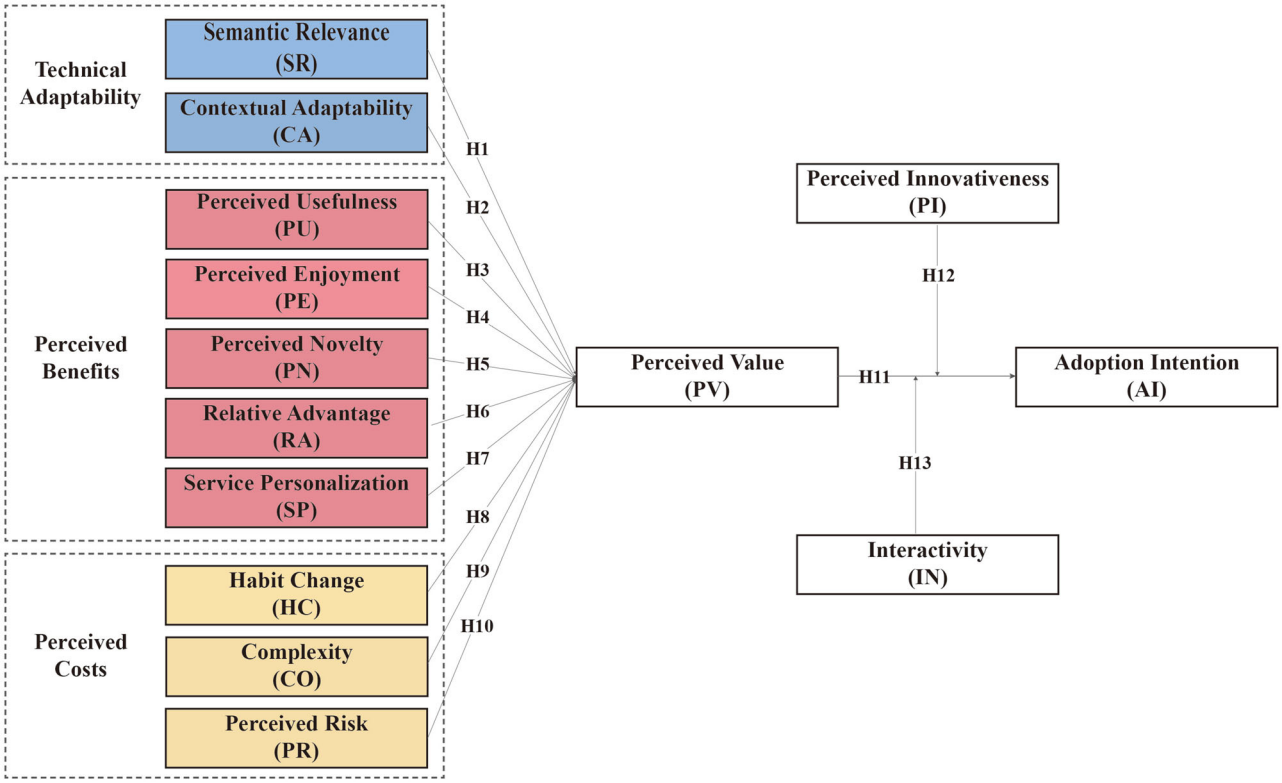

Fig. 1 | Proposed conceptual model.

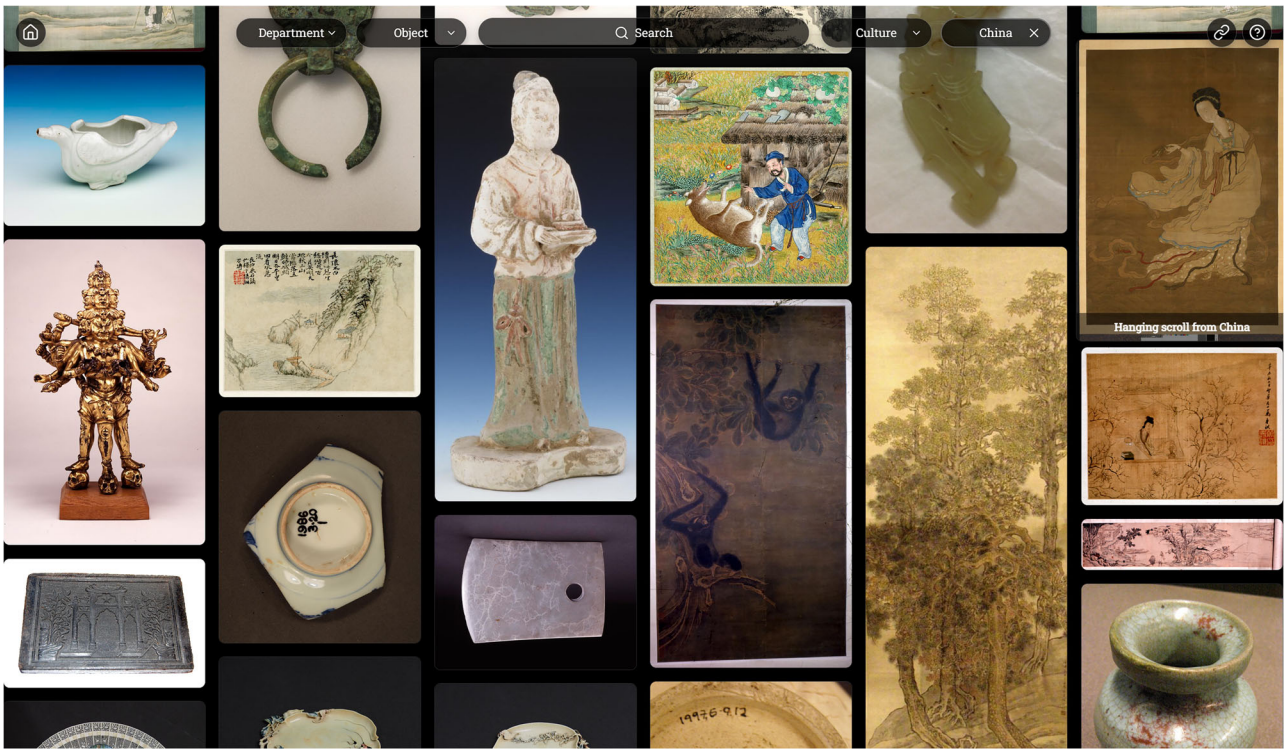

Fig. 2 | A representative usage scenario of The Living Museum platform.

established scales in the literature and modified to suit the specific context of GenAI interactions in museums. Table 3 presents each variable along with its corresponding source. As all original measurement items were published in English, a rigorous translation process was undertaken. Three professional translators with experience in academic research were engaged to translate and cross-proof the items through multiple iterative rounds. In addition, domain experts in artificial intelligence and museum studies were consulted to evaluate the clarity, structural coherence, and contextual appropriateness of the questionnaire. Their insights informed refinements to the language and structure to enhance content validity. Before launching

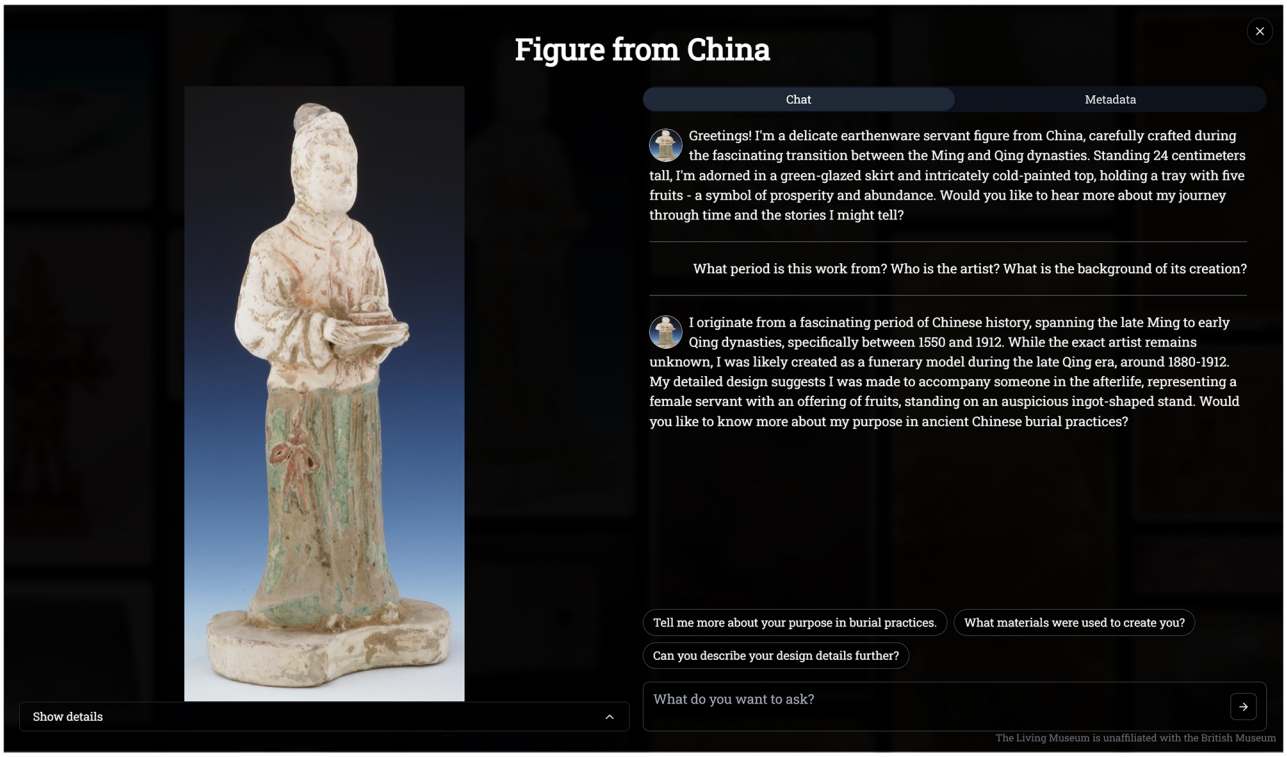

Fig. 3 | An example of GenAI-enabled interaction within The Living Museum.

the full-scale survey, a pilot test was conducted with 30 participants. Based on their feedback, further revisions were made to improve item clarity, user comprehension, and contextual relevance.

Data collection

Data for this study were collected via the Chinese online survey platform Wenjuanxing between 20 May and 17 June 2025. All respondents were residents of mainland China. According to Talmi (https://livingmuseum.app, accessed 1 June 2025), the objective of employing artificial intelligence language models in museums, exemplified by The Living Museum, is to facilitate interactive experiences that are both informative and engaging. To ensure diversity and enhance sample representativeness, a stratified random sampling approach was adopted. Participants were stratified along two key dimensions: occupational background and visit purpose. To further ensure data reliability, the questionnaire included a screening item: "Have you completed the 'Living Museum' experience?" Respondents who answered 'No' were screened out, and their responses were excluded from the dataset. A total of 798 responses were collected. After eliminating incomplete and duplicate entries, 726 valid responses remained, yielding a response rate of 91%.

Regarding the sample size, Partial Least Squares Structural Equation Modeling (PLS-SEM), as a variance-based structural equation modeling technique[78], imposes no strict upper limit on sample size. As long as the minimum sample requirement is met, it can accommodate datasets of varying scales[79,80]. According to the "10-times rule"[81], the sample size should be at least ten times the number of estimated parameters or the most complex latent variable indicators in the model to ensure stability and statistical power. Likewise, Kline[82] suggests that for complex structural models, a sample size of no fewer than 200 is necessary to maintain model robustness. In this study, we further referred to Clemente et al.[83] and Maccallum & Bryant[84] and conducted a power analysis using the G-Power 3.1 software to determine the minimum required sample size. Under the conditions of an effect size of $f^2 = 0.15$, a significance level of $\alpha = 0.05$, and a statistical power of $(1-\beta) = 0.80$, with ten predictors directed toward PV (Total predictors = 10; Tested predictors = 10), the analysis indicated a minimum required sample size of $N = 118$. The actual sample used in this study ($N = 726$) far exceeds this threshold, providing sufficient statistical power. Although previous studies have noted that an excessively large sample may cause even small effects to appear statistically significant[85], the relatively large sample in this study is necessary. Our model includes multiple latent variables, moderation effects, and multi-group comparisons across different user categories. A larger sample ensures the stability of parameter estimation and the robustness of group-level results. This approach is consistent with methodological recommendations from Cheah et al.[80] and Sarstedt et al.[86], who emphasized that increasing sample size is both reasonable and necessary when dealing with complex models or multi-group analyses. Finally, to mitigate the potential issue of "spurious significance" resulting from a large sample, the interpretation of results in this study goes beyond significance levels alone. We also report effect sizes and the substantive meaning of path coefficients to provide a balanced and rigorous evaluation. Taken together, both theoretical and methodological considerations confirm that the sample size in this study is fully justified and appropriate.

All participants provided informed consent prior to participation. The collected data are used exclusively for academic purposes and will not be disclosed for commercial use. All personally identifiable information is kept strictly confidential in accordance with standard ethical research protocols.

Sample Analysis

Among the 726 valid respondents, gender distribution was nearly balanced, with 370 males (51.0%) and 356 females (49.0%). The largest age group was 18–25 years, comprising 42.3% of the sample ($n = 307$), followed by respondents aged 26–35 (24.8%, $n = 180$), 36–45 (20.2%, $n = 147$), 46–55 (9.2%, $n = 67$), and those aged 55 and above (3.4%, $n = 25$). Regarding educational attainment, the majority of respondents (87.2%) held a bachelor's degree or higher. Specifically, 57.7% ($n = 419$) had a bachelor's degree, 19.3%($n = 140$) held a master's degree, and 10.2% ($n = 74$) had obtained a doctoral degree. The remaining 12.8%($n = 93$) had not completed undergraduate education. In terms of occupation, students represented the largest group (35.8%, $n = 260$), followed by cultural heritage

Table 3 | Questionnaire for Variable Items and Reference

Variable	Items	Issue	Reference
Semantic Relevance (SR) (3 items)	SR1	The AI's answers matched my questions.	Li et al.[33]
	SR2	The content aligned with my expectations.	
	SR3	The responses were focused and relevant.	
Contextual Adaptability (CA) (5 items)	CA1	The AI considered my visit purpose or background.	Jin et al.[114]
	CA2	The content adjusted to my question style or preferences.	
	CA3	The AI used a communication style familiar to me.	
	CA4	The content matched my interests.	
	CA5	The AI seemed aware of my context.	
Perceived Usefulness (PU) (4 items)	PU1	The AI helped me understand the exhibits.	Xu et al.[112]
	PU2	It improved my exploration efficiency.	
	PU3	It helped me find useful information.	
	PU4	The AI service supported my learning goals.	
Perceived Enjoyment (PE) (3 items)	PE1	I enjoyed using the AI features.	Kim et al.[23]
	PE2	Interacting with the AI was fun.	
	PE3	The AI experience increased my engagement.	
Perceived Novelty (PN) (3 items)	PN1	Using AI was a new experience for me.	Hameed et al.[106]
	PN2	The interaction felt unique and fresh.	
	PN3	It offered a new way to explore culture.	
Relative Advantage (RA) (3 items)	RA1	This platform was better than traditional tours.	Nguyen[113]
	RA2	It was more efficient than guides or labels.	
	RA3	It improved my overall visit experience.	
Service Personalization (SP) (4 items)	SP1	The platform recommended content I liked.	Seon et al.[111]
	SP2	The service felt tailored to me.	
	SP3	I could personalize my tour.	
	SP4	The AI adapted to my feedback.	
Habit Change (HC) (3 items)	HC1	It changed how I usually visit museums.	Skandali et al.[110]
	HC2	I had to adjust to using AI.	
	HC3	Adopting this service required effort.	
Complexity (CO) (3 items)	CO1	The AI features were hard to use.	Garrouch and Mzoughi[105]
	CO2	Learning the system was not easy.	
	CO3	It took time to interact with AI.	
Perceived Risk (PR)	PR1	I was concerned about data privacy.	Nguyen[113]
	PR2	I questioned the accuracy of the AI.	
	PR3	I worried it may harm traditional visits.	
Perceived Value (PV) (4 items)	PV1	The AI added value to my visit.	Kim et al.[23]
	PV2	Benefits outweighed my time and effort.	
	PV3	The experience was worth the time.	
	PV4	It was valuable for learning and exploring.	
Adoption Intention (AI) (4 items)	AI1	I would keep using this AI service.	Kim et al.[23]
	AI2	I plan to use it again.	
	AI3	I would recommend it to others.	
	AI4	I will include it in future visits.	
Perceived Innovativeness (PI) (3 items)	PI1	I'd try updated versions of the service.	Hameed et al.[106]
	PI2	I like trying new AI in museums.	
	PI3	I use cultural platforms with new tech.	
Interactivity (IN) (5 items)	IN1	The AI answered quickly and clearly.	Xu et al.[93]
	IN2	It felt like real conversation.	
	IN3	I could explore topics freely.	
	IN4	The interaction felt immersive.	
	IN5	The conversation was smooth and natural.	

researchers (24.5%, $n = 178$), museum professionals (19.4%, $n = 141$), teachers (12.1%, $n = 88$), and respondents from other fields (8.1%, $n = 59$). With respect to visit purpose, general exploration (e.g., browsing or gaining an overview) was the most common motivation (43.1%, $n = 313$), followed closely by in-depth research visits aimed at acquiring detailed knowledge of exhibits or professional content (41.9%, $n = 304$). Planned on-site visits for future travel or project preparation accounted for 15.0% ($n = 109$). A detailed demographic breakdown is provided in Table 4.

Measurement model

This study employed SPSS 27.0 and SmartPLS 4.0 to analyse the empirical data and assess the measurement model. Internal consistency reliability was first evaluated using Cronbach's alpha and composite reliability. As shown in Table 5, Cronbach's alpha values for all constructs ranged from 0.849 to 0.988, while CR values ranged from 0.921 to 0.990. Both sets of values exceed the recommended threshold of 0.70, indicating strong internal consistency across the measurement items.

Convergent validity was then assessed through the average variance extracted (AVE). All AVE values ranged from 0.796 to 0.961, surpassing the minimum acceptable value of 0.50. This suggests that a substantial proportion of variance is captured by the corresponding latent variables. Furthermore, all standardised factor loadings in the confirmatory measurement model ranged from 0.852 to 0.988, well above the 0.70 threshold, thereby providing further evidence of satisfactory convergent validity.

Discriminant validity was further assessed using the Fornell–Larcker[87] criterion. According to this method, a construct is considered to have discriminant validity if the square root of its average variance extracted (AVE) exceeds its correlations with all other constructs. As shown in Table 6, the square roots of the AVEs (presented along the main diagonal) were consistently greater than the corresponding inter-construct correlation coefficients, confirming the presence of discriminant validity. To complement this analysis, the heterotrait–monotrait ratio of correlations (HTMT) was also calculated. As presented in Table 7, all the HTMT values were below the 0.90 threshold[88]. These findings provide further support for the discriminant validity of the measurement model. Collectively, the results confirm that all constructs in the formal questionnaire exhibit satisfactory discriminant validity, thereby justifying their inclusion in the subsequent structural equation modelling analysis.

Evaluating the structural model

Following model estimation, the study evaluated its explanatory and predictive power using the coefficient of determination (R^2) and the predictive relevance statistic (Q^2). Multicollinearity was also assessed via the variance inflation factor (VIF), with results summarized in Table 8 and Table 9. Consistent with Marcoulides[89], R^2 values range from 0 to 1, with higher values indicating stronger explanatory power. According to Hair et al.[90]., Q^2 values greater than 0, 0.25, and 0.50 indicate low, medium, and high predictive relevance, respectively. In this study, the adjusted R^2 values were 0.408 for PV and 0.923 for Adoption Intention (AI), while the corresponding Q^2 values were 0.381 and 0.487. These results demonstrate that the model possesses substantial explanatory power and high predictive validity. Furthermore, the VIF values for all constructs in the structural model ranged from 1.441 to 1.884, well below the commonly accepted threshold of 3. This indicates that multicollinearity among the predictor variables is not a concern. Model fit was further assessed using the standardized root mean square

Table 4 | Statistical information on the demographic distribution of the participants, n = 726

Variable	Category	Frequency	Percent
Gender	Male	370	51.0
	Female	356	49.0
Age	18–25	307	42.3
	26–35	180	24.8
	36–45	147	20.2
	46–55	67	9.2
	Above55	25	3.4
Education	Below undergraduate	93	12.8
	Undergraduate	419	57.7
	Postgraduate	140	19.3
	Doctor	74	10.2
Occupation	Student	260	35.8
	Teacher	88	12.1
	Cultural Heritage Researcher	178	24.5
	Museum Professional	141	19.4
	Other	59	8.1
Purpose of Visit	General browsing	313	43.1
	Planning an on-site visit	109	15.0
	In-depth academic or research-oriented visit	304	41.9
Total		726	100

Table 5 | Reliability and validity analysis

Variable	Factor loading	Cronbach 's alpha	Composite reliability	AVE
Semantic Relevance(SR)	0.882–0.899	0.872	0.921	0.796
Contextual Adaptability(CA)	0.852–0.862	0.879	0.917	0.734
Perceived Usefulness(PU)	0.868–0.884	0.898	0.929	0.765
Perceived Enjoyment(PE)	0.877–0.892	0.859	0.914	0.780
Perceived Novelty(PN)	0.878–0.896	0.867	0.919	0.790
Relative Advantage(RA)	0.881–0.889	0.864	0.917	0.786
Service Personalization(SP)	0.856–0.886	0.895	0.927	0.761
Habit Change(HC)	0.873–0.891	0.856	0.912	0.776
Complexity(CO)	0.869–0.883	0.849	0.909	0.769
Perceived Risk(PR)	0.869–0.889	0.852	0.910	0.772
Perceived Value(PV)	0.869–0.898	0.908	0.936	0.784
Adoption Intention(AI)	0.938–0.949	0.957	0.969	0.887
Perceived Innovativeness(PI)	0.972–0.988	0.980	0.987	0.961
Interactivity(IN)	0.972–0.981	0.988	0.990	0.953

Table 6 | Discriminant validity (Fornell–Larcker criterion)

	AI	CA	CO	HC	IN	PE	PI	PN	PR	PU	PV	RA	SP	SR
AI	**0.942**													
CA	0.312	**0.856**												
CO	-0.365	-0.510	**0.877**											
HC	-0.440	-0.486	0.450	**0.881**										
IN	0.544	0.134	-0.057	0.022	**0.976**									
PE	0.419	0.467	-0.390	-0.555	0.012	**0.883**								
PI	0.609	-0.060	0.011	-0.092	-0.880	0.048	**0.980**							
PN	0.311	0.473	-0.462	-0.440	0.128	0.432	-0.066	**0.889**						
PR	-0.302	-0.399	0.353	0.376	-0.067	-0.367	0.015	-0.394	**0.879**					
PU	0.375	0.467	-0.388	-0.398	0.060	0.467	0.013	0.471	-0.393	**0.875**				
PV	0.328	0.481	-0.460	-0.416	0.552	0.455	-0.417	0.482	-0.411	0.458	**0.886**			
RA	0.204	0.403	-0.428	-0.353	0.135	0.307	-0.115	0.440	-0.419	0.335	0.412	**0.887**		
SP	0.419	0.458	-0.446	-0.574	-0.034	0.564	0.082	0.425	-0.366	0.379	0.412	0.351	**0.873**	
SR	0.330	0.450	-0.416	-0.430	0.087	0.428	-0.033	0.442	-0.374	0.467	0.450	0.363	0.467	**0.892**

Table 7 | Discriminant validity (HTMT values)

	AI	CA	CO	HC	IN	PE	PI	PN	PR	PU	PV	RA	SP	SR
AI	-													
CA	0.340	-												
CO	0.405	0.590	-											
HC	0.486	0.561	0.528	-										
IN	0.557	0.143	0.064	0.036	-									
PE	0.462	0.536	0.457	0.648	0.036	-								
PI	0.629	0.063	0.026	0.101	0.892	0.052	-							
PN	0.342	0.542	0.539	0.511	0.140	0.500	0.071	-						
PR	0.334	0.461	0.415	0.440	0.074	0.429	0.021	0.458	-					
PU	0.405	0.526	0.444	0.454	0.070	0.532	0.040	0.534	0.449	-				
PV	0.350	0.536	0.523	0.471	0.587	0.514	0.444	0.543	0.467	0.506	-			
RA	0.224	0.461	0.499	0.411	0.147	0.356	0.124	0.510	0.488	0.379	0.465	-		
SP	0.453	0.515	0.512	0.657	0.042	0.643	0.088	0.483	0.418	0.423	0.455	0.398	-	
SR	0.362	0.513	0.485	0.498	0.093	0.494	0.044	0.510	0.434	0.527	0.503	0.418	0.529	-

Table 8 | Values of R^2 and Q^2

Variable	R-square	R-square adjusted	Q-square predict
Perceived Value(PV)	0.416	0.408	0.381
Adoption Intention(AI)	0.924	0.923	0.487

Table 9 | Model fit indices and Common method bias

Fit Index	Computed Values	Threshold	Reference
SRMR	0.031	≤0.08	Henseler et al.[115]
NFI	0.854	≥0.80	Gupta and Singharia[116]
VIF	1.441 - 1.884	<3	Hair et al.[90]

residual (SRMR) and the normed fit index (NFI). The SRMR value was 0.031, and the NFI value was 0.854, both indicating a satisfactory model fit.

Structural model and hypothesis test

To evaluate the structural relationships among variables, the research model was assessed using partial least squares (PLS) algorithms, two-tailed bootstrapping (5000 resamples, $\alpha = 0.05$), and blindfolding procedures.

As presented in Table 10 and illustrated in Fig. 4, both SR ($\beta = 0.104$, $t = 2.070$, $p < 0.05$) and CA ($\beta = 0.112$, $t = 2.161$, $p < 0.05$) exhibited significant positive effects on PV, thereby supporting hypotheses H1 and H2. Regarding perceived benefits, PU ($\beta = 0.111$, $t = 2.173$, $p < 0.05$), PE ($\beta = 0.123$, $t = 2.144$, $p < 0.05$), PN ($\beta = 0.129$, $t = 2.518$, $p < 0.05$), and RA ($\beta = 0.096$, $t = 2.176$, $p < 0.05$) were all found to significantly enhance PV, supporting hypotheses H3, H4, H5 and H6. However, SP ($\beta = 0.014$, $t = 0.273$, $p = 0.785$) did not exert a significant influence on PV, and thus hypothesis H7 was not supported. With respect to perceived costs, both CO ($\beta = -0.122$, $t = 2.461$, $p < 0.05$) and PR ($\beta = -0.094$, $t = 2.194$, $p < 0.05$) had significant negative effects on PV, supporting hypotheses H9 and H10. Conversely, HC ($\beta = -0.015$, $t = 0.287$, $p = 0.774$) was not significantly associated with PV, and hypothesis H8 was therefore not supported. Moreover, PV ($\beta = 0.864$, $t = 30.040$, $p < 0.001$) had a strong positive effect on Adoption Intention (AI), lending support to hypothesis H11.

In the moderation path analysis, both PI ($\beta = 0.462$, $t = 14.396$, $p < 0.001$) and IN ($\beta = 0.340$, $t = 8.729$, $p < 0.001$) showed significant moderating effects on the path from PV to AI. Thus, Hypotheses H12 and H13 are

supported. To clarify these interaction effects, a simple slope diagram was constructed (Fig. 5) to illustrate how the impact of PV on AI varies across different levels of the moderators.

Post hoc analyses: multi-group analysis (MGA)

To further investigate how user background influences the mechanisms underlying GenAI adoption intentions, a multi-group analysis was conducted. The sample was categorised into two groups: non-professional users ($N = 407$) and professional users ($N = 319$). The non-professional group included students ($n = 260$), teachers ($n = 88$), and participants from other fields ($n = 59$), whereas the professional group was composed of cultural heritage researchers ($n = 178$) and museum professionals ($n = 141$).Using the multi-group analysis function in SmartPLS 4.0, we compared the structural path coefficients between the two groups, focusing on statistically significant differences ($p < 0.05$).

As shown in Table 11, the influence of perceived novelty (PN) on perceived value (PV) differed significantly between the groups ($p = 0.011 < 0.05$). For non-professional users, this effect was not significant ($\beta = 0.029$, $p = 0.691$). In contrast, among professional users, perceived novelty exerted a strong positive effect on PV ($\beta = 0.296$, $p < 0.001$). Likewise, perceived risk (PR) demonstrated a statistically significant difference in its effect on PV across groups ($p = 0.022 < 0.05$), suggesting a context-dependent evaluation of risk. For non-professional users, perceived risk had a significantly negative effect on PV ($\beta = -0.158$, $p = 0.003$), whereas for professional users, the effect was not significant ($\beta = 0.044$, $p = 0.534$).

Table 10 | Hypotheses tests

H	Paths	β	M	SD	T	P	Results
H1	SR → PV	0.104	0.105	0.05	2.070	0.039	Supported
H2	CA → PV	0.112	0.111	0.052	2.161	0.031	Supported
H3	PU → PV	0.111	0.111	0.051	2.173	0.030	Supported
H4	PE → PV	0.123	0.123	0.057	2.144	0.032	Supported
H5	PN → PV	0.129	0.128	0.051	2.518	0.012	Supported
H6	RA → PV	0.096	0.097	0.044	2.176	0.030	Supported
H7	SP → PV	0.014	0.016	0.053	0.273	0.785	Not supported
H8	HC → PV	-0.015	-0.014	0.053	0.287	0.774	Not supported
H9	CO → PV	-0.122	-0.122	0.049	2.461	0.014	Supported
H10	PR → PV	-0.094	-0.094	0.043	2.194	0.028	Supported
H11	PV → AI	0.864	0.865	0.029	30.040	0.000	Supported
H12	PI x PV → AI	0.462	0.462	0.032	14.396	0.000	Supported
H13	IN x PV → AI	0.340	0.340	0.039	8.729	0.000	Supported

SR semantic relevance, *CA* contextual adaptability, *PU* perceived usefulness, *PE* perceived enjoyment, *PN* perceived novelty, *RA* relative advantage, *SP* service personalization, *HC* habit change, *CO* complexity, *PR* perceived risk, *PV* perceived value, *AI* adoption intention, *PI* perceived innovativeness, *IN* interactivity.

Discussion

This study applies the Value-Based Adoption Model (VAM) together with a GenAI adaptability framework to explain how users perceive the value of GenAI in museums and what shapes their adoption intentions.

First, the significant effects of semantic relevance (SR) and contextual adaptability (CA) on perceived value (PV) (H1–H2) highlight the distinctive demands of cultural heritage dissemination. Unlike general digital contexts, museum visitors are highly sensitive to accuracy and contextual relevance[1]. When AI outputs align semantically with collection narratives and visitor queries, users' trust in museums as authoritative knowledge sources is reinforced[7], thereby strengthening value perceptions. Dynamic contextual responsiveness, such as adapting to visit objectives, prior knowledge, or linguistic style, further enhances narrative coherence and knowledge transfer[91]. This supports Ivanov's claim that when technology adapts actively to user states, instrumental and experiential value reinforce each other[35]. GenAI's adaptability thus

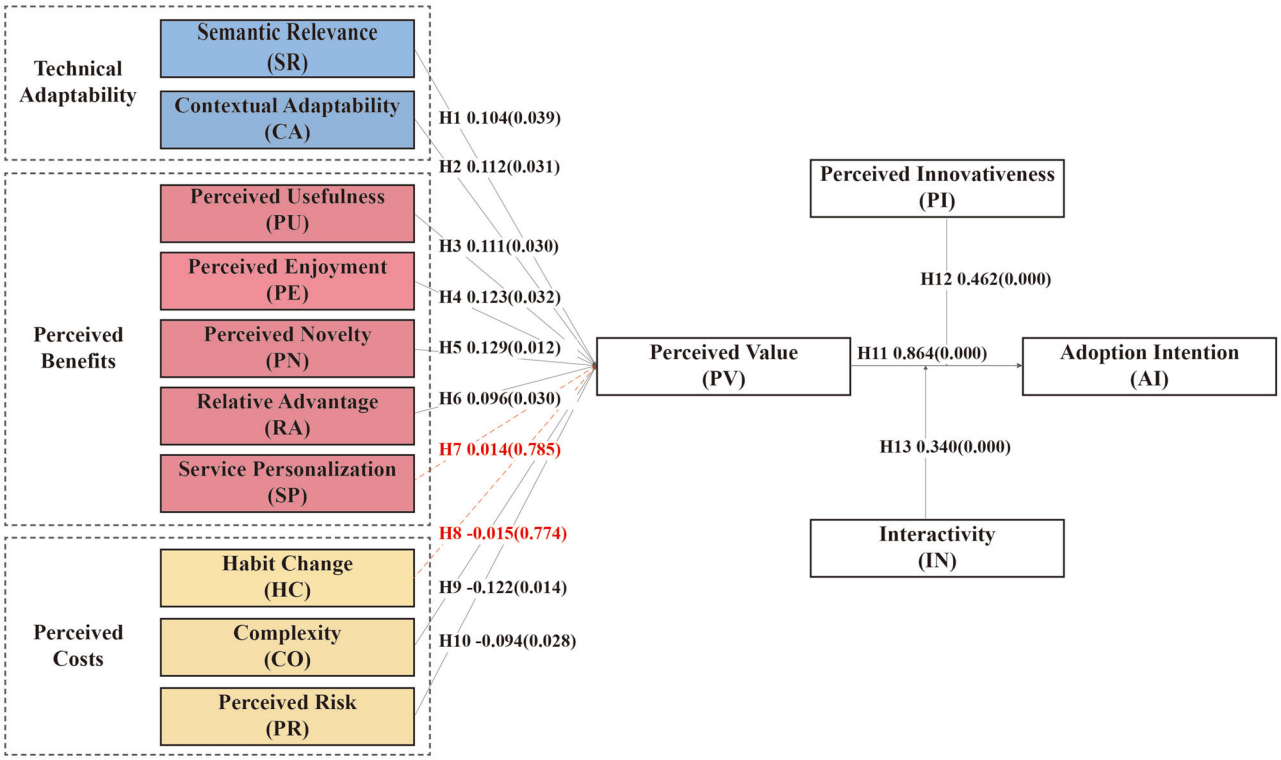

Fig. 4 | Results of research model test.

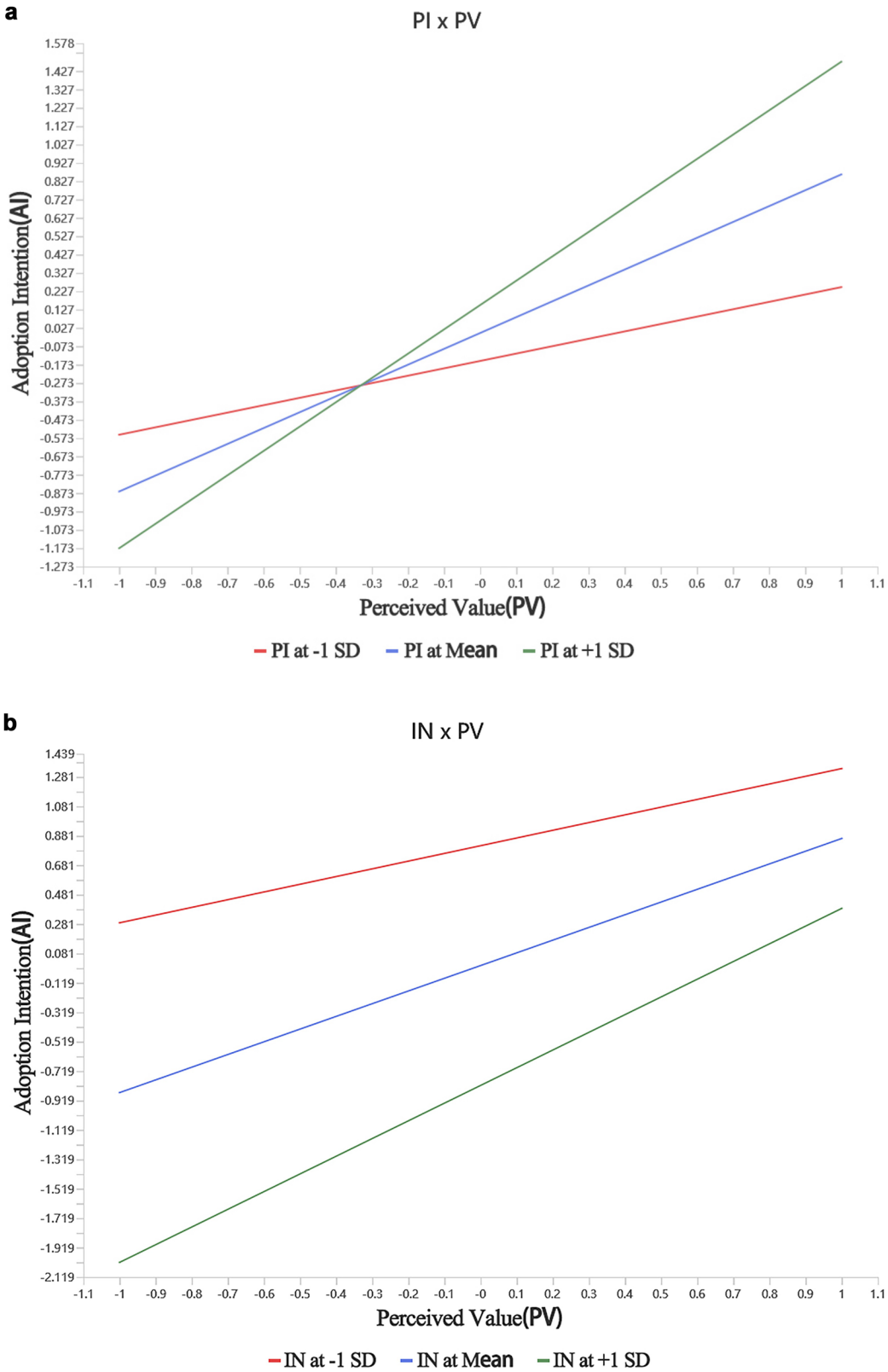

Fig. 5 | Moderating effects of Perceived Innovativeness(PI) and Interactivity(IN) on the Perceived Value (PV)–Adoption Intention (AI) relationship.
a PI×PV → AI: When PI is one standard deviation(+ 1 SD) above the mean, PV exerts its strongest positive effect on AI. At average PI levels, the effect remains moderately strong, while at low PI levels (-1 SD) below the mean), the effect is substantially weaker. These results indicate that PI positively moderates the PV–AI relationship. **b** IN×PV → AI: Although the interaction term is overall significant and positive, the simple slope analysis shows a different pattern. PV has its strongest positive effect on AI when IN is low (-1 SD). The effect remains moderately strong at average IN levels, but becomes weaker when IN is high (+ 1 SD). This suggests that while the moderating effect of IN is positive overall, it diminishes locally. In highly interactive contexts, the marginal contribution of PV to AI is progressively reduced.

Table 11 | Multi-Group analysis by occupation

Paths	Non-professional User (n = 407)		Professional User (n = 319)		P (Difference)	Results
	β	P	β	P		
SR → PV	0.128	0.059	0.070	0.348	0.565	No
CA → PV	0.178	0.008	0.066	0.427	0.292	No
PU → PV	0.071	0.282	0.162	0.041	0.371	No
PE → PV	0.153	0.040	0.042	0.649	0.352	No
PN → PV	0.029	0.691	0.296	0.000	0.011	Significant
RA → PV	0.089	0.115	0.105	0.122	0.856	No
SP → PV	-0.034	0.609	0.089	0.273	0.239	No
HC → PV	-0.036	0.547	-0.007	0.933	0.783	No
CO → PV	-0.123	0.056	-0.108	0.155	0.881	No
PR → PV	-0.158	0.003	0.044	0.534	0.022	Significant
PV → AI	0.866	0.000	0.862	0.000	0.941	No
PI → AI	0.123	0.022	0.190	0.000	0.371	No
IN → AI	-0.816	0.000	-0.811	0.000	0.956	No
IN x PV → AI	0.386	0.000	0.288	0.000	0.214	No
PI x PV → AI	0.478	0.000	0.445	0.000	0.620	No

SR semantic relevance, *CA* contextual adaptability, *PU* perceived usefulness, *PE* perceived enjoyment, *PN* perceived novelty, *RA* relative advantage, *SP* service personalization, *HC* habit change, *CO* complexity, *PR* perceived risk, *PV* perceived value, *AI* adoption intention, *PI* perceived innovativeness, *IN* interactivity.

enables "high relevance, low cognitive load", which enhances cultural value transmission and PV.

Second, in terms of perceived benefits, the positive influence of perceived usefulness (PU), perceived enjoyment (PE), perceived novelty (PN), and relative advantage (RA) on PV (H3–H6) indicates that museum users pursue dual objectives: functional efficiency and emotional fulfillment. PU and RA reflect users' practical needs for GenAI, while PE and PN capture emotional engagement, consistent with prior findings[44]. By contrast, service personalisation (SP) showed no significant effect (H7), diverging from expectations and from most prior studies[10]. This may be because The Living Museum platform already incorporates default personalisation, such as automatic matching of exhibition content with queries, blurring users' distinction between system adaptation and explicit personalisation. In addition, within cultural heritage contexts, users may prioritise the authority of AI-generated content over the freedom to customise[11]. The weak role of SP may therefore reflect tension between algorithmic recommendations and users' preference for autonomous cultural exploration.

In terms of perceived costs, the negative effects of complexity (CO) and perceived risk (PR) on PV (H9–H10) underscore challenges in applying GenAI to cultural heritage. CO increases operational and cognitive burden, interrupting immersive experiences when users feel fatigued[92]. PR encompasses privacy concerns as well as doubts about the reliability and cultural appropriateness of AI-generated content. Because museums function as authoritative domains of knowledge, any perceived inaccuracy or bias can rapidly erode PV[24]. Meanwhile, habit change (HC) had no significant effect (H8). This may reflect the sample composition: 67.1% of respondents were under 35, a group typically more adaptive to new technologies[93]. Behavioural shifts are therefore not viewed as costly, particularly since virtual and physical visits follow similar pathways, lowering behavioural transition barriers.

Third, the strong predictive effect of PV on adoption intention (AI) (H11) confirms the relevance of value-based theories in cultural heritage. Users quickly form cost–benefit judgements of "worth using," which translate into adoption intentions, consistent with VAM[23,30] and related studies[94–96]. However, this pathway is moderated by perceived innovativeness (PI) and interactivity (IN) (H12–H13). Innovative users are more willing to experiment with GenAI, and when they perceive value, they are more likely to adopt, consistent with IDT[97]. The moderating effect of IN is more nuanced: while positive overall, it weakens under conditions of high interactivity. Once systems already provide fluid and immersive interactions, additional gains in PV from more information or functionality become limited. This finding challenges the assumption that "more interactivity is always better" highlighting the need to balance functionality with experiential coherence.

Finally, multi-group analysis revealed professional differences. Among cultural heritage researchers and museum staff, perceived novelty had a stronger positive effect on value, suggesting that knowledge-driven users value GenAI's capacity for new forms of interaction and expression. Among non-professional users, however, perceived risk exerted a stronger negative effect. Concerns about accuracy and security, therefore, weigh more heavily on their value judgements. These differences reflect structural variations in error tolerance, verification capacity, and institutional trust (https://bit.ly/43VAzq0, accessed 12 August 2025). Professionals can better evaluate and correct model errors, whereas non-professionals rely more on perceived credibility and institutional authority.

This study contributes meaningfully to the theoretical development of digital cultural heritage and technology adoption research. First, it proposes and empirically validates an integrated conceptual model linking the adaptability of GenAI, operationalized through SR and CA, with PV and subsequent behavioral intention to adopt. This framework extends user experience theory by incorporating AI-specific dimensions into the context of cultural heritage digitization. Second, the study systematically integrates both benefit-driven constructs, such as perceived usefulness, enjoyment, novelty, relative advantage, and service personalization, and cost-related constructs, including habit change, complexity, and perceived risk. The empirical results support the core tenet of value-based decision-making theory: users access technologies by weighing expected benefits against anticipated costs. Specifically, perceived usefulness, enjoyment, novelty, and relative advantage significantly enhance PV, whereas complexity and perceived risk exert a detrimental effect. Third, the findings highlight the moderating roles of perceived innovativeness and interactivity. These attributes amplify the positive relationship between PV and adoption intention, particularly in digitally mediated cultural settings. These findings underscore the need to design GenAI systems that extend beyond basic functionality, offering features that spark curiosity and foster interactive, responsive user experiences. Within this broader analytical framework, the multi-group analysis suggests distinct differences in how various user groups engage with GenAI. Professional users, such as museum staff and cultural heritage researchers, typically value technological novelty and innovation. In contrast, non-professional users place greater emphasis on usability, intuitive design, and minimizing perceived risks. These findings suggest that users' professional backgrounds play a substantial role in shaping their expectations and the criteria they use to evaluate GenAI technologies.

At the practical level, this study provides specific recommendations for museum managers and AI developers. First, semantic relevance and contextual adaptation should form the foundation of GenAI systems. To minimize semantic drift, systems should draw on museum-validated knowledge bases, glossaries, or knowledge graphs. They should also include functions, such as intent recognition, adaptive adjustment of language and reading difficulty, and location or gallery-based prompts. These measures ensure that AI responses remain aligned with the user's immediate tasks. Second, in terms of perceived benefits, experience-enhancing features can be layered onto the visit without disrupting its flow. Examples include modules, such as "key artifact highlights", "comparative tracing", and "academic citation or further reading". These features enable users to access accurate information more quickly and accomplish specific goals. The rhythm of interaction can be managed by introducing multimodal presentations, visualizations, or micro-

interactions at critical points. Such design helps prevent novelty-driven distractions that may reduce attention or interrupt learning. Third, regarding perceived costs, interface and process design should adopt standardized defaults, such as example-as-prompt, one-tap initiation, and reversible errors to reduce complexity. Advanced functions should follow the principle of progressive disclosure and be supported by reusable query templates and explanations of terminology, thereby lowering the threshold for effective use. To mitigate perceived risk, AI outputs should provide embedded source attributions, confidence indicators, and verifiable evidence. When uncertainty arises, systems should supply transparent explanations that specify both reasoning and limitations, while also offering access to human assistance or expert validation. Fourth, strategies should reflect the moderating effects of perceived innovativeness (PI) and interactivity (IN). For users with high PI, functions, such as "trial of new features" and "update visibility," can reduce barriers to turning value judgments into active engagement. For interactivity, which is inherently complex, adaptive mechanisms are recommended. These include switchable quiet or high-density modes and controls for prompt frequency and session length, which help prevent over-interaction. Finally, user heterogeneity must be addressed. For professional users, "research mode" and advanced functions should be placed in prominent positions to foster deeper exploration and emphasize novelty. For non-professional users, a simplified default process should be retained, with greater emphasis on credibility cues and clear privacy or security notices. This ensures an adaptive AI experience that accommodates diverse user needs.

This study offers empirical evidence on the application of GenAI in museums and the mechanisms shaping PV and adoption willingness. Nonetheless, several limitations must be acknowledged. First, the research is based on a single online museum platform (The Living Museum) and does not assess GenAI performance in offline exhibition contexts. As a result, the operational dynamics of technological adaptability, perceived benefits, and perceived costs in physical settings require further empirical validation. In addition, the sample primarily consists of users from mainland China, raising concerns about the generalizability of the findings across cultural contexts, museum types, and broader populations. Future studies should therefore investigate real-world or hybrid environments and adopt multi-site sampling across museums and regions to test the model's cross-cultural robustness. Second, this study relies largely on cross-sectional self-reported questionnaire data, which may be subject to social desirability bias and common method bias. Although PLS-SEM is appropriate for predictive and exploratory modelling, research reliability and causal inference would benefit from methodological diversification. Longitudinal tracking, field experiments, and behavioural log analysis could strengthen validity. Complementary approaches, such as covariance-based SEM for confirmatory testing, Bayesian SEM, or causal diagramming, could further assess model sensitivity. Third, while this study considers core antecedents of perceived benefits and perceived costs, as well as moderators of adoption intention, important mediators and boundary conditions may remain unexamined. Moreover, the key variables are primarily measured through subjective perceptions, which are shaped by cognitive abilities, levels of technological literacy, and short-term experiential impressions. Future research should therefore incorporate additional latent variables, such as trust, willingness to pay, sense of control, and social influence. Combining subjective measures with objective or expert-based assessments—such as expert fact-checking of GenAI outputs or automated semantic similarity metrics—could provide stronger validation and enhance reliability.

Data Availability

The datasets used and/or analyzed during the current study are available from the corresponding author (Junping Xu) upon reasonable request.

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Author contributions

X.H.: Writing original draft, Methodology, Investigation, Data curation, Software. J.X.: Validation, Supervision, Writing review and editing. Y.W.: Formal analysis, Visualization.

Competing interests

The authors declare no competing interests.

Additional information

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